Step-by-Step Assembly (LEGO-style)

Build the monster by following the pictures, just like a LEGO booklet. Do the steps **in order**. In every picture the new part is inside a **yellow dashed box**, and the parts from earlier steps stay where they are.

> **Important:** keep the power **unplugged** while you build. Plug it in only at Step 7.

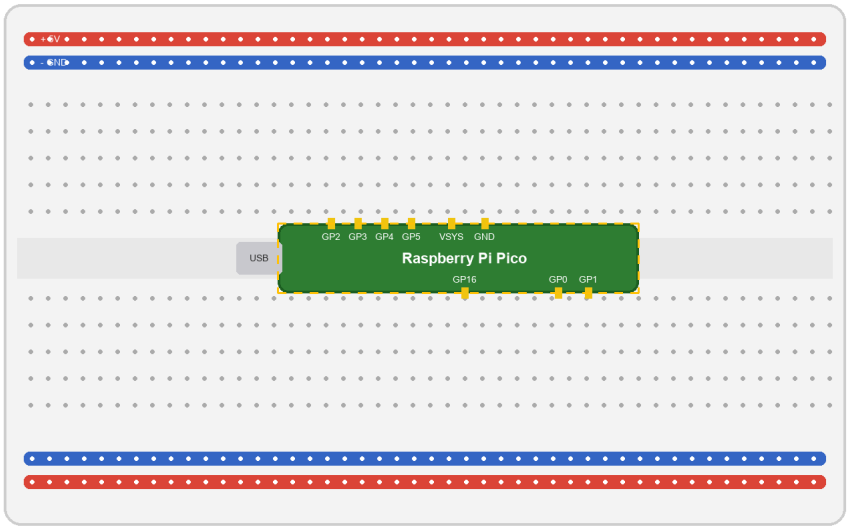
What you need

Part	How many
Raspberry Pi Pico	1
Distance sensor	2
Servo motor (with the head)	1
MP3 player + SD card	1
Speaker	1
5V power supply	1
Breadboard	1
Jumper wires (red, black, and colours)	a bunch

The two long lines on the breadboard are the power lines: the **red** line is `+5V` and the **blue** line is `GND` (the minus). All the holes in one line are joined together.

Step 1 — Put the brain on the breadboard

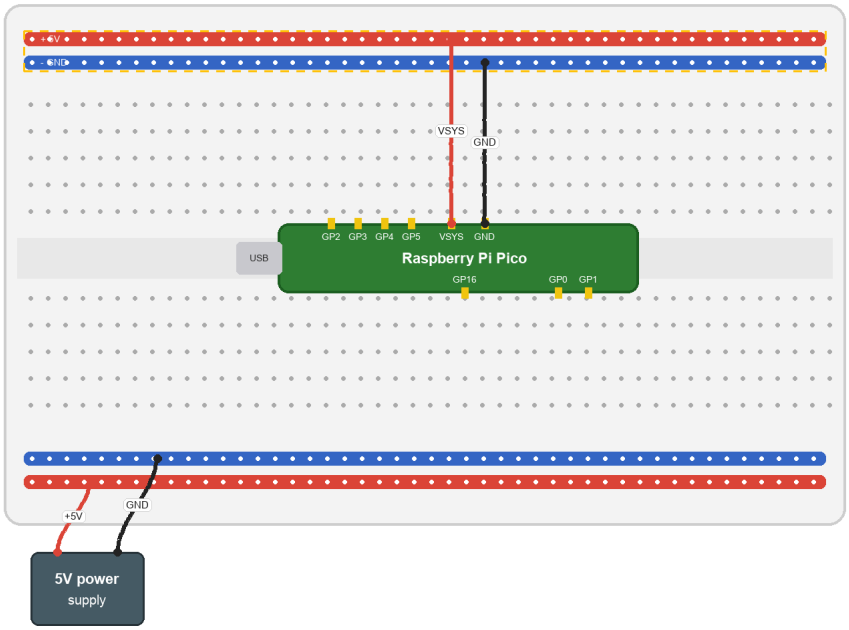
Step 1 - Place the Pico on the breadboard



Push the Raspberry Pi Pico into the middle of the breadboard so it sits across the centre gap. The little USB plug points to the left.

Step 2 — Give it power

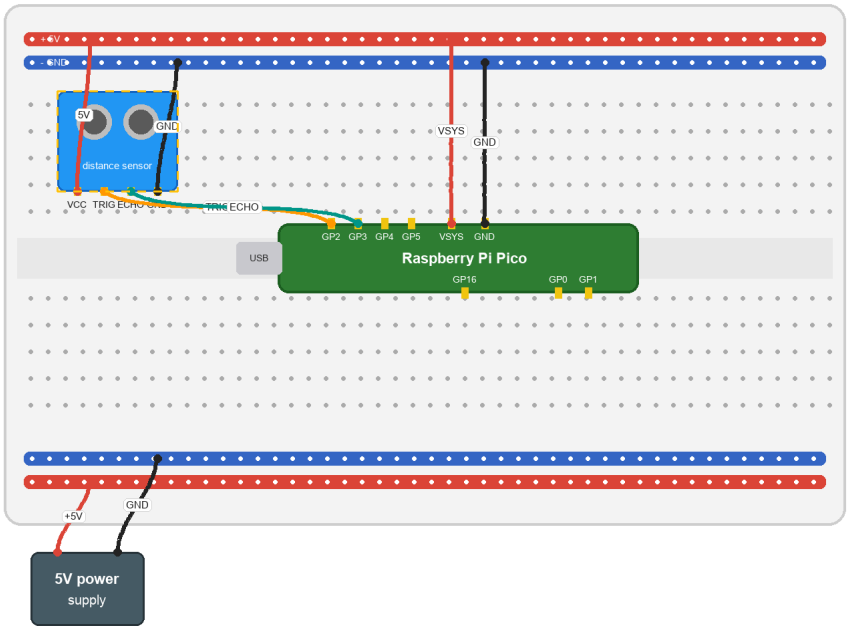
Step 2 - Connect the 5V power rails



Connect the 5V power supply to the red and blue lines. Then run two wires from the power lines to the Pico: red to **VSYS** and black to **GND**.

Step 3 — Add the left eye

Step 3 - Add the LEFT distance sensor

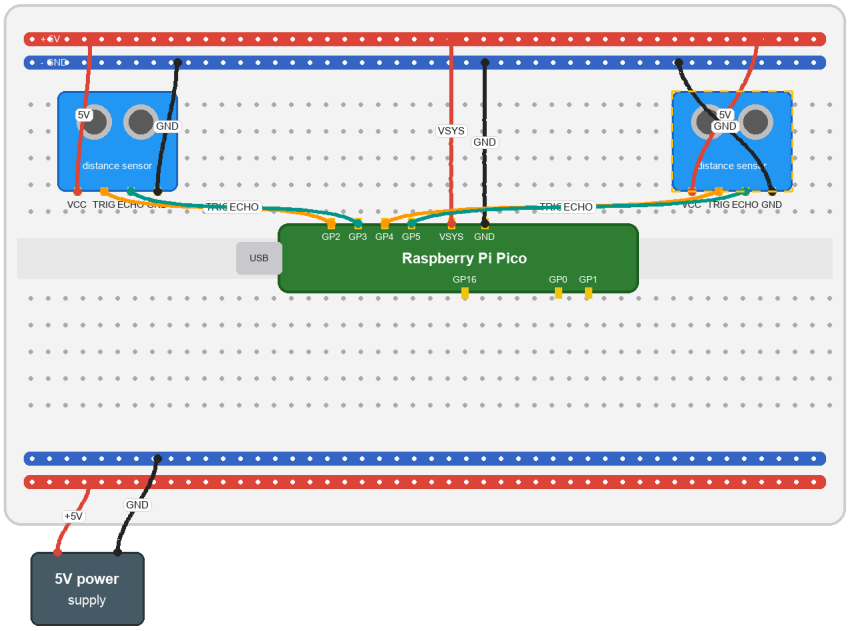


Put the left distance sensor at the top-left. Connect it like this:

Sensor leg	Goes to
VCC	red line (+5V)
GND	blue line (GND)
TRIG	Pico **GP2**
ECHO	Pico **GP3**

Step 4 — Add the right eye

Step 4 - Add the RIGHT distance sensor

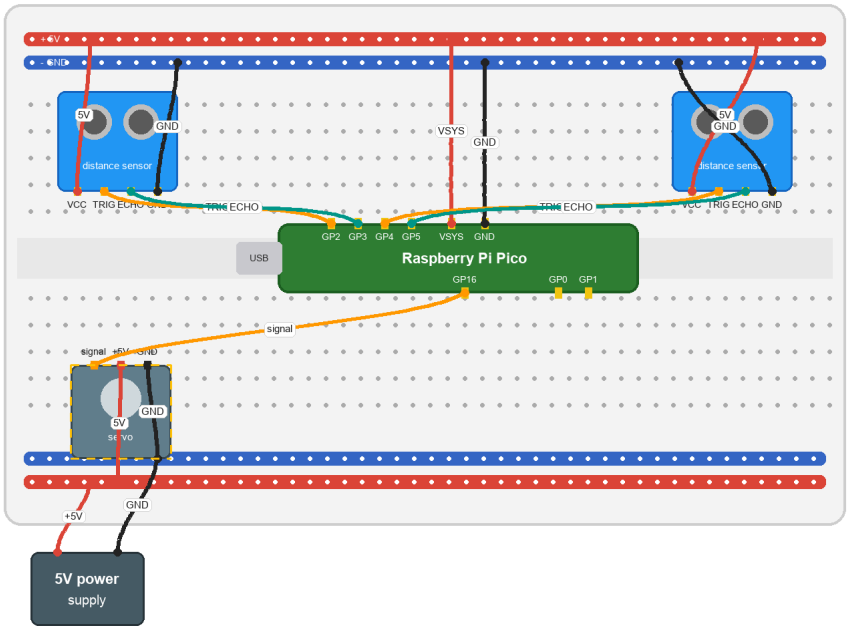


Put the right distance sensor at the top-right, the same way. This time the wires go to different Pico legs:

Sensor leg	Goes to
VCC	red line (+5V)
GND	blue line (GND)
TRIG	Pico **GP4**
ECHO	Pico **GP5**

Step 5 — Add the neck

Step 5 - Add the head servo

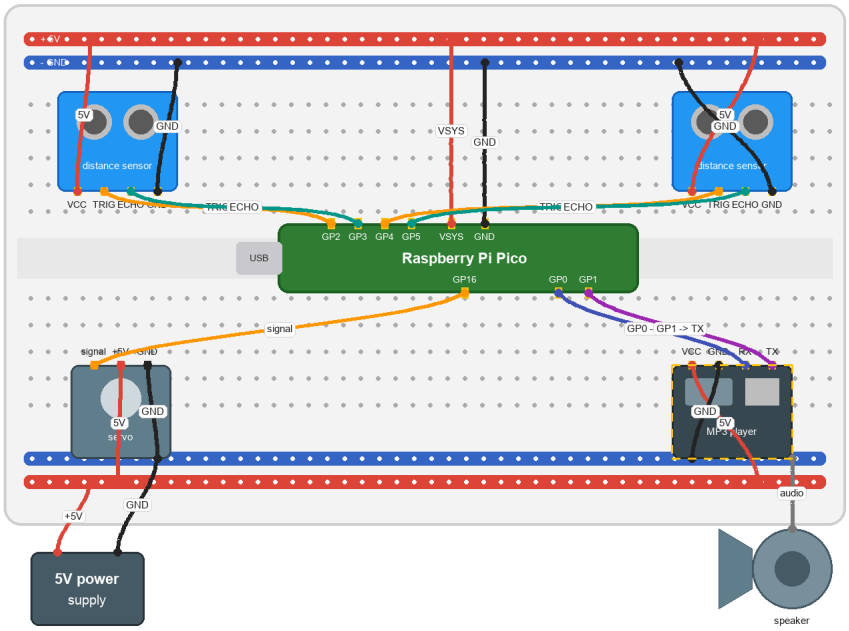


Plug in the servo. Its three wires go to the Pico and the power lines:

Servo wire	Goes to
signal (orange)	Pico **GP16**
+5V (red)	red line (+5V)
GND (brown or black)	blue line (GND)

Step 6 — Add the voice

Step 6 - Add the MP3 player and speaker



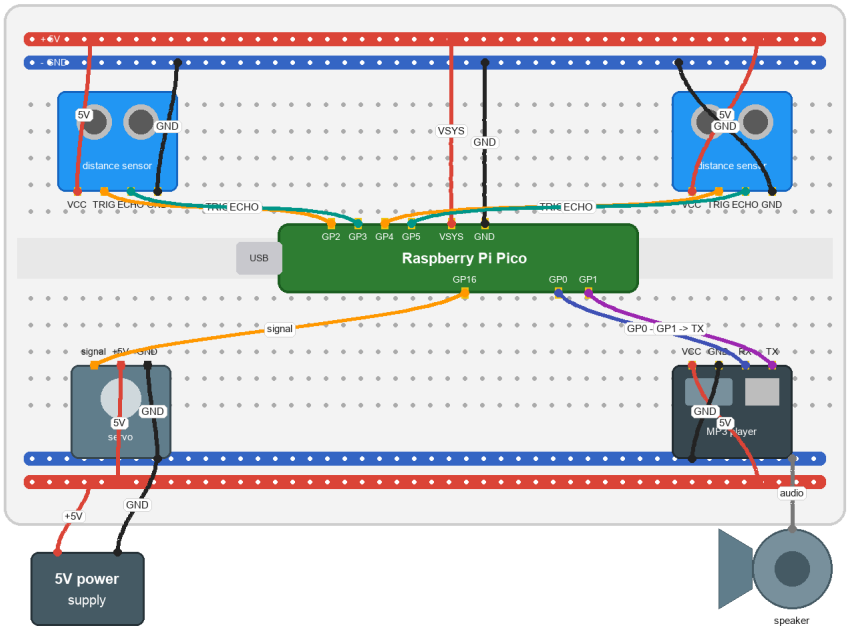
Connect the MP3 player at the bottom-right and plug in the speaker. The two data wires **cross over**: the Pico sends on GP0 to the player's RX, and listens on GP1 from the player's TX.

MP3 player pin	Goes to
VCC	red line (+5V)
GND	blue line (GND)
RX	Pico **GP0**
TX	Pico **GP1**

Put the scary song on the SD card as the **first track**, then put the card in the player.

Step 7 — The finished monster

Step 7 - The finished monster



Look at the picture one last time. Every part must share the same **GND** (the blue line). Now plug in the power and test:

1. Wave your hand in front of the eyes. The head should follow you.
2. Move closer than 30 cm. The scary song should start.
3. Move away. The head should look straight ahead again.

If it does not work

Problem	What to check
Nothing happens	Is the power plugged in? Red to red, black to blue?
The head does not move	Is the signal wire on GP16? Does the servo have 5V?
The head shakes	The servo needs more power; use a stronger 5V supply and share the GND.
No sound	Is the SD card in? Is the song the first track? Are TX and RX crossed?
It follows the wrong way	Swap the two sensors' TRIG and ECHO wires.

Strange behaviour	Every GND must meet at the blue line (common ground).